SWINBURNE UNIVERSITY OF TECHNOLOGY

# Exploratory Data Analysis Report
# Emotion Detection Dataset

Facial Recognition with Emotion and Liveness
COS30082 Applied Machine Learning, Group Project

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**Source notebook:** emotion_detection_eda.ipynb

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**Data:** Google Drive folder “Emotion Detection” (train, test)

# 1. Summary

This report describes the image dataset that our team will use to train the emotion detection model. Before training, we checked *what is in the data*: how many images each emotion has, whether the files are clean, whether the classes are balanced, and whether the test data accidentally contains pictures the model has already seen. All numbers come from running the project notebook on the real dataset.

| Question | Answer | Verdict |
|---|---|---|
| How big is the dataset? | 7 emotions; 28,721 training and 7,178 test images; no validation split | Large enough |
| Are the files clean? | No corrupted files. All images are 48x48 greyscale PNG | Good |
| Is it balanced? | No. Happy has 7,215 images, disgusted only 436 (ratio 16.5). Gini 0.25 | Imbalanced |
| Is image quality good? | About 2% of scanned images are flagged (mostly too bright); a few are completely black or blank | Acceptable |
| Do emotions look different? | Yes on average (average faces differ clearly), but many single images are ambiguous | Good, with noise |
| Is there data leakage or label noise? | 24 identical and 26 near-identical pictures appear in both train and test; 5 groups of identical pictures carry two different emotion labels | Needs action |

> **Main message**
> The data is clean and usable, but three issues need a team decision before training: (1) the **class imbalance** (disgusted is only 1.5% of the data), (2) **leakage** of a few dozen images from train into test, and (3) a small amount of **label noise** (blank images and identical pictures labelled with different emotions).

> **Important limit of this run**
> The quality scan opened a random **20%** of the images (7,180 of 35,899), because reading files straight from Google Drive is slow. Percentages are reliable estimates; counts of duplicates and leaked images are **minimum** values. A full scan (using a zip copy) will give exact numbers.

# 2. What data we have and how we checked it

The shared folder has two parts, *train* and *test*, and each contains one sub-folder per emotion: angry, disgusted, fearful, happy, neutral, sad and surprised. The folder name is the label. Every image is a small 48x48 greyscale face. There is no official validation split, so we created one from the training data (Section 6). Unlike the face recognition data, no person identities are given, so we cannot check whether the same person appears in train and test; we can only compare pictures.




Figure 1. Random training images. Captions show the emotion and its numeric label.

## How the checks work

| Check | Plain-language meaning | How it is measured |
|---|---|---|
| Size / format | Are all images the same shape and file type? | Width, height, file type and colour mode |
| Sharpness | Is the picture blurry? | Variance of the Laplacian (an edge detector). Low = blurry |
| Brightness / contrast | Too dark, too bright, or washed out? | Average and spread of grey levels (0 to 255) |
| Duplicates | Is the same picture stored twice? | MD5 hash = identical file; perceptual hash = near copy |
| Label conflict | Is one picture filed under two emotions? | Duplicate groups that contain more than one emotion |
| Leakage | Does the test data contain training pictures? | Compare hashes between train and test |

# 3. Class distribution

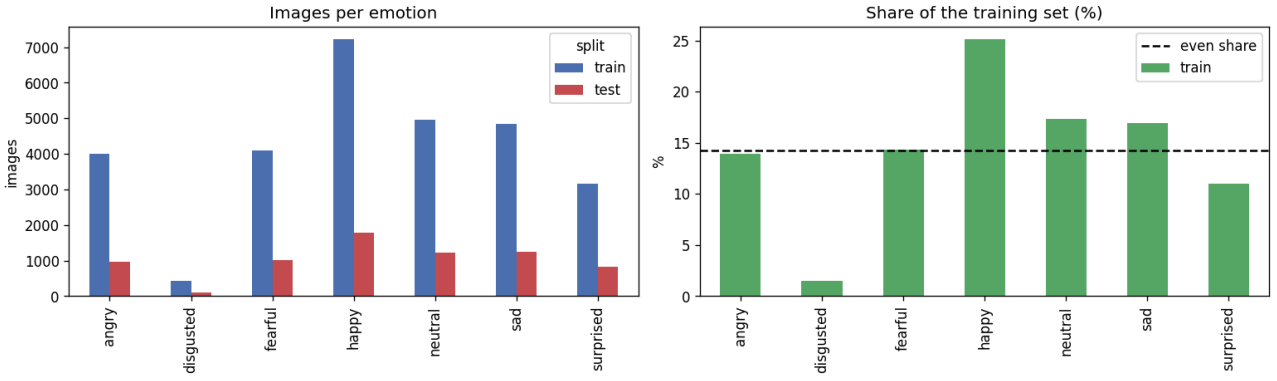



Figure 2. Images per emotion in each split (left) and share of the training set (right). The dashed line is the even share of 14.3%.

| Emotion | Train | Test | Train share (%) | Class weight |
|---|---|---|---|---|
| angry | 3,992 | 958 | 13.9 | 1.028 |
| disgusted | 436 | 111 | 1.5 | 9.411 |
| fearful | 4,097 | 1,024 | 14.3 | 1.001 |
| happy | 7,215 | 1,774 | 25.1 | 0.569 |
| neutral | 4,966 | 1,233 | 17.3 | 0.826 |
| sad | 4,844 | 1,247 | 16.9 | 0.847 |
| surprised | 3,171 | 831 | 11.0 | 1.294 |
| **Total** | **28,721** | **7,178** | **100** | |

The split between train and test is consistent: each emotion has almost the same share in both (for example happy 25.1% and 24.7%). The classes themselves, however, are far from even. **Happy** holds a quarter of the data, while **disgusted** has only 436 training images (1.5%), against an even share of 14.3%. The imbalance ratio (largest class divided by smallest) is 16.5, and the Gini coefficient is 0.25 (0 means perfectly even, 1 means very uneven).

> **What this means for training**
> A model can reach about 25% accuracy by always answering “happy”, and it can ignore disgusted almost completely while still looking good on average accuracy. We should use **class weights** (the table above; disgusted gets about 9.4 times the weight of an average class) or balanced sampling, and report **macro-F1 and a confusion matrix**, not only accuracy.

# 4. Image properties and quality

## 4.1 Size, format and quality flags

All 7,180 scanned images are 48x48 PNG files in single-channel greyscale (mode L), so no resizing or colour conversion is needed for a 48x48 network. For networks that expect larger or 3-channel input, images must be up-scaled and the channel repeated, which cannot add detail.

| Flag | Train | Test | Meaning |
|---|---|---|---|
| Corrupted | 0 | 0 | File cannot be opened |
| Low resolution (<40 px) | 0 | 0 | Image too small |
| Blurry (threshold 30) | 5 | 0 | Very soft or blank images |
| Too dark (mean <40) | 19 | 6 | Nearly black images |
| Too bright (mean >200) | 76 | 23 | Over-exposed or white background |
| Low contrast (std <20) | 14 | 6 | Flat, washed-out images |
| Images scanned | 5,707 | 1,473 | 20% random sample |

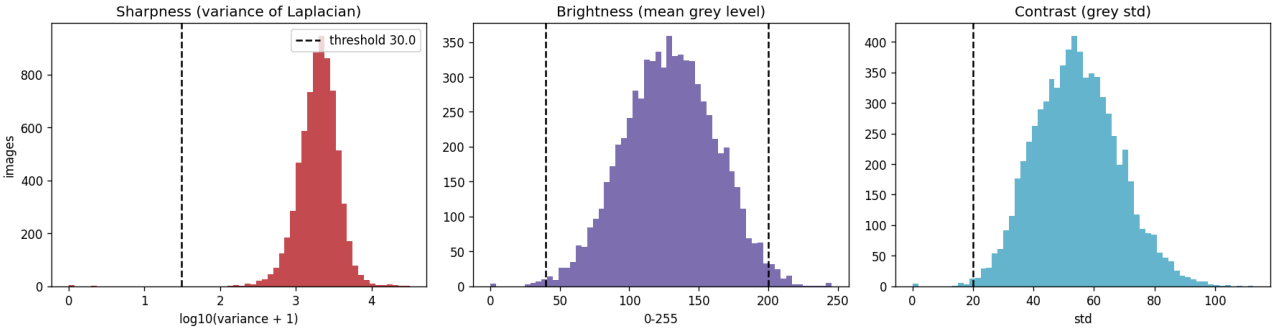



Figure 3. Distribution of sharpness, brightness and contrast. Dashed lines are the thresholds used to raise a flag.

The three histograms are smooth and single-peaked. Brightness is centred near 128 with small tails at both ends, which are the dark and bright flags. Only about 2% of the scanned images raise any flag, so the data is generally clean.

## 4.2 Looking at the extreme images

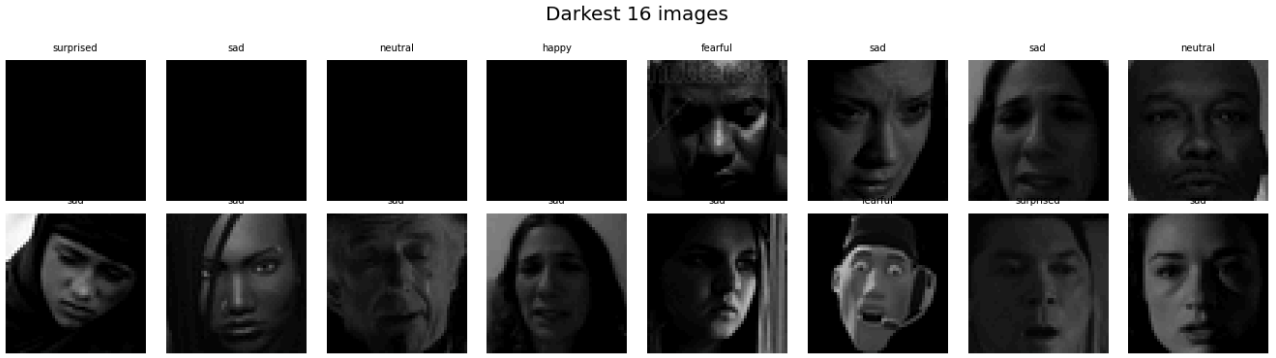



Figure 4. The 16 darkest images. The first four are completely black (no face at all); the others are real faces in very low light.




Figure 5. The 24 blurriest images. Several are blank or empty, and the others are soft but still recognisable faces.

> **Finding 1: blank images exist**
> Figures 4 and 5 contain completely black and completely white images with an emotion label. They carry no information and can only add noise. They should be removed (they are easy to find: contrast is close to 0).

**Finding 2: the blur threshold is too low**
Normal images have a sharpness near 2,000 (the peak in Figure 3 is at about 10^3.3), while the starting threshold is 30. Only 5 images were flagged. The 1st percentile of sharpness is 419, so a threshold of roughly 400 to 750 would flag the worst 1 to 5%. The team should choose it by inspecting grids like Figure 5, and apply it to **training data only**.

# 5. Appearance of each emotion

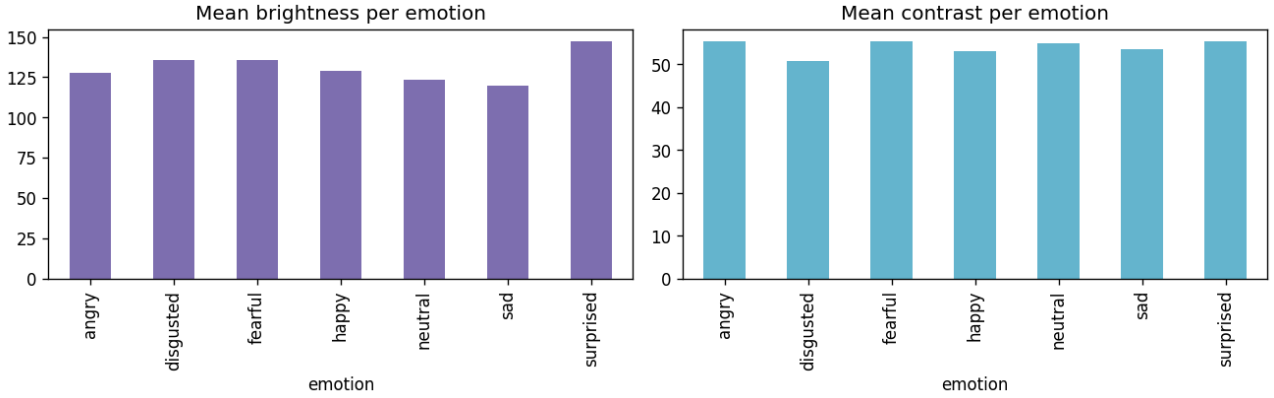



Figure 6. Average brightness (left) and contrast (right) per emotion.

| Emotion | Brightness | Contrast | Sharpness |
|---|---|---|---|
| angry | 127.5 | 55.3 | 2,222 |
| disgusted | 135.9 | 50.7 | 1,617 |
| fearful | 135.4 | 55.3 | 2,170 |
| happy | 128.7 | 53.0 | 1,981 |
| neutral | 123.7 | 54.9 | 1,968 |
| sad | 120.0 | 53.5 | 1,788 |
| surprised | 147.4 | 55.4 | 2,444 |

Brightness ranges from 120 (sad) to 147 (surprised), and contrast from 51 (disgusted) to 55 (surprised). The differences are small compared with the spread inside each class, but a model could pick up a weak lighting shortcut (for example, brighter images tend to be surprised). Brightness and contrast augmentation reduces this risk. Disgusted images are slightly less sharp because the class is small and many pictures were enlarged from a lower resolution.




Figure 7. Random training faces, one row per emotion.

Within each row, the faces vary strongly in age, gender, skin tone, head pose, glasses and hands covering the face. Some examples look like a different emotion from their label (for example a laughing face under disgusted, or a screaming face under sad). This is typical for emotion datasets collected from the web, where labels are given by crowd workers and are not always agreed.

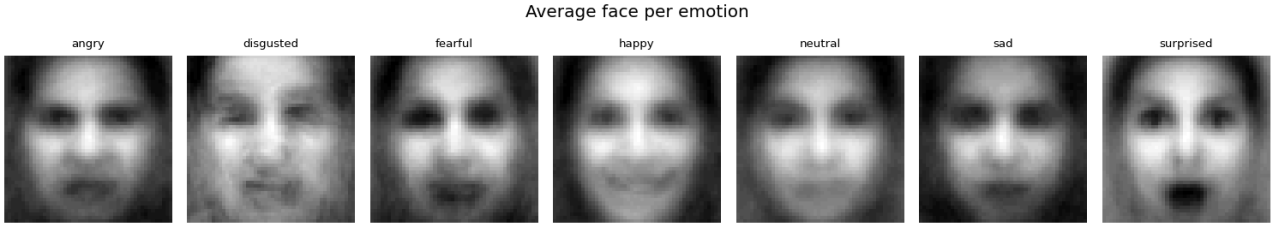



Figure 8. Average face of each emotion.

The average faces show that the classes *are* visually distinct: surprised has a wide open mouth, happy shows a smile, angry and sad have lowered eyebrows, and disgusted is blurrier because its images are the most varied and the class is the smallest. Fearful, neutral and sad look similar, so we should expect the model to confuse them.

# 6. Duplicates, label conflicts and leakage

| Duplicate groups | Exact (MD5) | Near (pHash) |
|---|---|---|
| Inside one emotion | 57 | 5 |
| Across train and test | 23 | 2 |
| Different emotions (label conflict) | 5 | 0 |
| **Total** | **85** | **7** |

A **duplicate group** is a set of images that are identical (exact) or look identical (near). Most groups are harmless copies inside the same emotion, but two kinds are a problem.

## 6.1 Label conflicts




Figure 9. Identical pictures that carry different emotion labels. The captions show the labels given to each copy.

Five groups of identical pictures are filed under two or more emotions (Figure 9: the same face labelled sad, angry and surprised, or fearful and sad). Several of these are blank black images. These labels cannot all be right, so the images should be reviewed and removed or relabelled.

## 6.2 Leakage between train and test

| Compared groups | Identical images | Near copies | Verdict |
|---|---|---|---|
| train vs test | 24 | 26 | **Leakage** |

**Data leakage** means that an image used to test the model was also used to train it, so the score looks better than it truly is. 24 identical and 26 near-identical images are shared by train and test in the 20% sample. The full dataset will contain more (about five times as many if the sample is representative), but the share is still small, roughly 1% of the test set.

> **Why this matters**
> The leaked images are few, so the effect on accuracy is small, but it is easy to fix: delete from the test set (or from train) every image whose hash matches the other split. Note that the test set must stay fixed once the team has agreed on it, so decide before reporting any results.

### 6.3 Validation split

| Emotion | Train | Validation | Test |
|---|---|---|---|
| angry | 3,593 | 399 | 958 |
| disgusted | 392 | 44 | 111 |
| fearful | 3,687 | 410 | 1,024 |
| happy | 6,493 | 722 | 1,774 |
| neutral | 4,469 | 497 | 1,233 |
| sad | 4,360 | 484 | 1,247 |
| surprised | 2,854 | 317 | 831 |

The dataset has no validation set, so the notebook takes 10% of each emotion from the training data (2,873 images, random seed 42) to tune the model. This is a *stratified* split, so every emotion keeps the same proportion. The result is stored in *emotion_inventory_with_val.csv*. Because duplicates exist inside the training data, the cleaner approach is to remove duplicate groups first and then split.

## 7. Recommendations

| # | Action | Priority | Who decides |
|---|---|---|---|
| 1 | Run the notebook once on a zip copy of the dataset (SCAN_FRACTION = 1.0) to get exact duplicate, leakage and quality counts | High | Data team |
| 2 | Remove blank / non-face images (very low contrast, completely black or white) | High | Data team |
| 3 | Review the 5 label-conflict groups; remove or relabel them | High | Team agreement |
| 4 | Remove test images that also appear in train (or the training copy) | High | All |
| 5 | Use class weights from class_weights.csv or balanced sampling (imbalance ratio 16.5) | High | Model team |
| 6 | Report macro-F1 and a confusion matrix, not only accuracy; never tune on the test set | High | Model team |
| 7 | Use the validation split from emotion_inventory_with_val.csv | High | All |
| 8 | Raise the blur threshold from 30 to about 400 to 750 after checking the grid (training data only) | Medium | Data team |
| 9 | Use brightness / contrast augmentation, small rotations and horizontal flips | Medium | Model team |
| 10 | Expect noisy labels and confusion between similar emotions (fearful / surprised, sad / neutral) | Info | All |

## 8. Limitations

(1) Only 20% of images were scanned, so duplicate and leakage counts are lower bounds. (2) Labels come from the dataset and were not verified by us; some are clearly ambiguous. (3) Age, gender and ethnicity are not labelled and were not inferred, so fairness across groups cannot be assessed from this analysis. (4) Quality thresholds are starting values chosen by us and should be confirmed by the team. (5) Images are small (48x48), so fine details such as eye or mouth shape are limited.

# Glossary

| Term | Simple explanation |
|---|---|
| Class / emotion label | One of the 7 emotions the model must predict |
| Imbalance ratio | Size of the biggest class divided by the smallest |
| Gini coefficient | 0 = every class has the same number of images; 1 = very uneven |
| Class weight | Extra importance given to rare classes in the loss function |
| Hash (MD5) | Short fingerprint of a file; identical files have identical fingerprints |
| Perceptual hash | Fingerprint of how an image looks; survives resizing and re-saving |
| Laplacian variance | Sharpness score; low means blurry |
| Label conflict | The same picture stored under two different emotions |
| Data leakage | Test images that the model has already seen during training |
| Macro-F1 | Average of the F1 score of every class; treats rare classes equally |
| Stratified split | A split that keeps each class in the same proportion |

Generated from the project notebook outputs (EDA_report.html, EDA_summary.md and CSV tables in the team Drive folder).